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Thermodynamic evaluation of the germanium-lutetium and dysprosium-germanium binary systems

Khadija Achgar*, Najim Selhaoui, Abdellah Iddaoudi, Rkia Tamim, Hassan Azza, Lahcen Bouirden

Laboratory of Thermodynamics and Energy (L. T. E.), Department of Physics, Faculty of Sciences, University Ibn Zohr, PO 8106, Hay Dakhla, 80000 Agadir, Morocco

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ABSTRACT

The Ge-Lu and Dy-Ge binary systems have been optimized using the available experimental data on phase diagrams and thermodynamic functions by means of the CALPHAD approach. The Ge-Lu system contains seven phases based on six intermetallic compounds: $Ge_{1.8}Lu$, Ge_3Lu_2, Ge_5Lu_4, $Ge_{10}Lu_{11}$, Ge_4Lu_5, and Ge_3Lu_5, which were all treated as stoichiometric compounds. For the Dy-Ge system seven intermetallic compounds were found: Dy_5Ge_3, Dy_5Ge_4, DyGe, Dy_2Ge_3, Dy_3Ge_5, $DyGe_2$ and $DyGe_3$. These phases were stoichiometric except Dy_5Ge_3 that has a homogeneity range, was treated by a two-sublattice model with substitution in each sublattice. A solution model has been used for the description of the excess term of the Gibbs energy for the liquid phase and the solid solution for the Dy-Ge system. The Ge-Lu system was examined using two different descriptions of the liquid phase: exponential dependence of temperature (Kaptay model) and linear dependence (Redlich-Kister model). The calculations based on the thermodynamic modeling are in good agreement with the phase diagram data and experimental thermodynamic values available in the literature. The process of modeling allows us to calculate the phase equilibria and thermodynamic properties of the liquid phase and intermetallic compounds for the first time. The results obtained are reported in a series of figures and tables in this manuscript.

1. Introduction

Compounds of rare earth elements (REEs) with germanium are amongst several alloys that have attracted considerable attention in recent years. These alloys are indispensable in different engineering applications due to their unique properties and their wealth of structures that individually pertain to covalent, ion, or metal crystals. They are especially attractive candidates for the chemical industry, metallurgy, optical engineering, and microelectronics [1]. It is also known that Germanium and its alloys have super- and semiconducting properties and can be used as resistive and magnetic materials [2,3].

On the other hand the (REEs) and their alloys and compounds with other elements provide inexhaustible sources of new materials. In general, there are two industries reliant on REEs: mature sectors like catalysts, glassmaking, lighting, and metallurgy; and secondly, emerging sectors including battery alloys, ceramics, and permanent magnets.

In order to satisfy the ever increasing needs for reliable thermodynamic data for this class of materials, to refine the REE-Ge phase diagrams and to improve germanium alloy production methods, it is important to investigate the phase diagrams and develop a thermodynamic database of these systems based on the CALPHAD method.

The current work is a continuation of similar studies of REE-Ge (REE = Y [4], Ho [5], Yb [6], Er [7] and Tm [8]) already calculated in our laboratory, and the first in thermodynamic modeling of Ge-Lu and Dy-Ge binary systems. The objectives of the present work are to calculate the phase diagram and the values of thermodynamic functions of REE-Ge (REE = Lu, Dy) systems in standard conditions, to compile the experimental information, and to study its consistency in comparison with the calculated results obtained by using the Redlich-Kister [9] and Kaptay [10–12] models.

2. Bibliographic description of the binary systems

2.1. Equilibrium diagram

2.1.1. The Ge-Lu system

The germanium-lutetium equilibrium phase diagram was established experimentally by Eremenko et al. [13] over the whole composition range. They used X-ray diffraction, metallography, and differential thermal analysis techniques. From there information Moffatt [14] and Massalski [15] construct the phase diagram in Fig. 1.

According to these authors [13] the Ge-Lu system is characterized

* Corresponding author.
E-mail address: achgar9008@gmail.com (K. Achgar).

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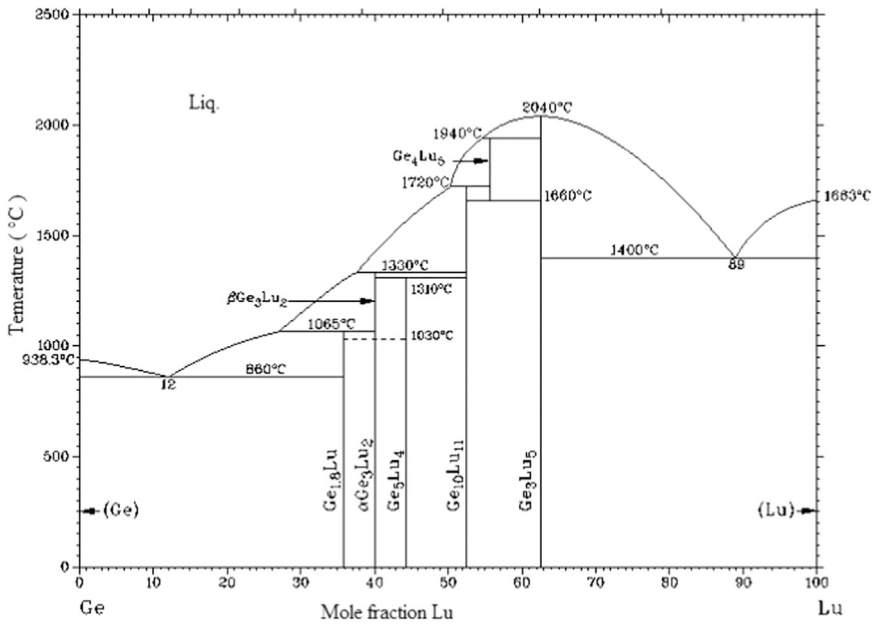

Fig. 1. Germanium-Lutetium phase diagram according to Eremenko et al. [13].

by:

- Ge_3Lu_5, the richest in Lu, is the most refractory compound in the Lu-Ge system and the only in this system that melts congruently at 2040 °C.
- Four compounds form peritectically Ge_4Lu_5 at 1940 °C, $Ge_{10}Lu_{11}$ at 1720 °C, βGe_3Lu_2 at 1330 °C, and $Ge_{1.8}Lu$ at 1065 °C.
- Ge_5Lu_4 decomposes peritectoidally at 1310 °C.
- Ge_3Lu_2 exhibits an allotropic transformation at 1030 °C; the crystal structure of the low temperature form is hexagonal, and the high-temperature modification is unknown
- Two eutectic reactions occur between (Ge) and $Ge_{1.8}$ Lu at 860 °C and 12 at% Lu and between Lu and Ge_3Lu_5 at 1400 °C and 89 at% Lu.
- The mutual terminal solid solubility of the pure elements appears to be very small.
- All the intermediate phases are stoichiometric.

2.1.2. The Dy-Ge system

The dysprosium-germanium phase equilibria have been determined by Eremenko et al. [16] Fig. 2 includes the following phases: liquid; pure elements (αDy), (βDy), and (Ge); and seven intermediate compounds.

- Dy_5Ge_3 melts congruently at 1825 °C and show a homogeneity region at about 1 at%.
- $DyGe_{1.63}$ forms during the peritectoid reaction and disintegrates into two solid phases βDy_2Ge_3 and DyGe2 at 935 °C.
- Dy_5Ge_4, DyGe, γDy_2Ge_3, $DyGe_2$, and $DyGe_{2.84}$ formed by peritectic reactions at 1575, 1530, 1405, 965, and 885 °C, respectively.
- The addition of germanium to dysprosium results in eutectic reaction among liquid, (αDy) and Dy_5Ge_3 at 1215 °C about 11 at% Ge in Dy rich part.
- The addition of dysprosium to germanium results in a eutectic reaction among liquid, $DyGe_{2.84}$ and (Ge) at 860 °C and about 85 at% Ge.
- The mutual solid-state solubility of dysprosium and germanium is lower than 1 at%.
- The Ge_3Lu_2 exhibits two allotropic transformations $\alpha Dy_2Ge_3 \leftrightarrow \beta Dy_2Ge_3 \leftrightarrow \gamma Dy_2Ge_3$ at 890 and 1300 °C, respectively.

2.2. Thermodynamic data

2.2.1. The Ge-Lu system

Meshel and Kleppa determined the standard enthalpies of formation of Ge_3Lu_5 by the mean of direct synthesis calorimetry at 1473 ± 2 K in two separate works [17,18]. The experimental values are − 73.5 ± 1.5 KJ mol^{-1} at^{-1} [17] and − 93.1 ± 2.2 KJ mol^{-1} at^{-1} [18]. In this work we take into account the recent one [18] because it seem to correlate with the data of the germanides of the earlier REE, contrarily to the value given in ref [17] which was significantly less exothermic. Gorbachuk and Bolgar [19] and Gorbachuk et al. [20] give the measured enthalpies H-H298 of Lu_5Ge_3 and Lu_2Ge_3, respectively. For Lu_5Ge_3 these quantities are measured by the method of mixtures with an error of 1.5% in the range 300–1300 K by means of a high-temperature differential calorimeter [21], while above 1300 K, it was measured with a high-temperature vacuum calorimeter [22]. Concerning Lu_2Ge_3 the experimental data [20] were corrected to allow for the impurity phase and fitted with the Meier–Kelly equation over a range from 442 to 1576 K: $H^\circ(T) - H^\circ(298.15\,K) = AT^2 + BT + CT^{-1} + D$. The coefficients of this equation were calculated with the least-squares method. Nikolaenko et al. [23] measured the integral enthalpy of mixing of liquid alloy for x(Lu) = 0.16 at 1275 K, the experimental value is − 29 KJ mol^{-1}.

2.2.2. The Dy-Ge system

Based on their calorimetric measurement at 1473 ± 2 K, Meshel and Kleppa [24] obtained the standard enthalpy of formation of Dy_5Ge_3 and equal to −92.4 ± 2.3 KJ mol^{-1} at^{-1}. Gorbachuk and Bolgar [25] determined the enthalpies of three dysprosium germanides Dy_5Ge_3, DyGe, and $DyGe_2$ between room temperature and the melting point. The enthalpies were examined by the method of mixtures with an error of not more than 1.5% with a high temperature differential calorimeter (DC) at 300–1200 K and with a high-temperature vacuum calorimeter (VC) above 1200 K [22,26].

2.3. Crystallographic data

The crystallographic data for the intermediate phases formed in these systems are reported in Tables 1 and 2.

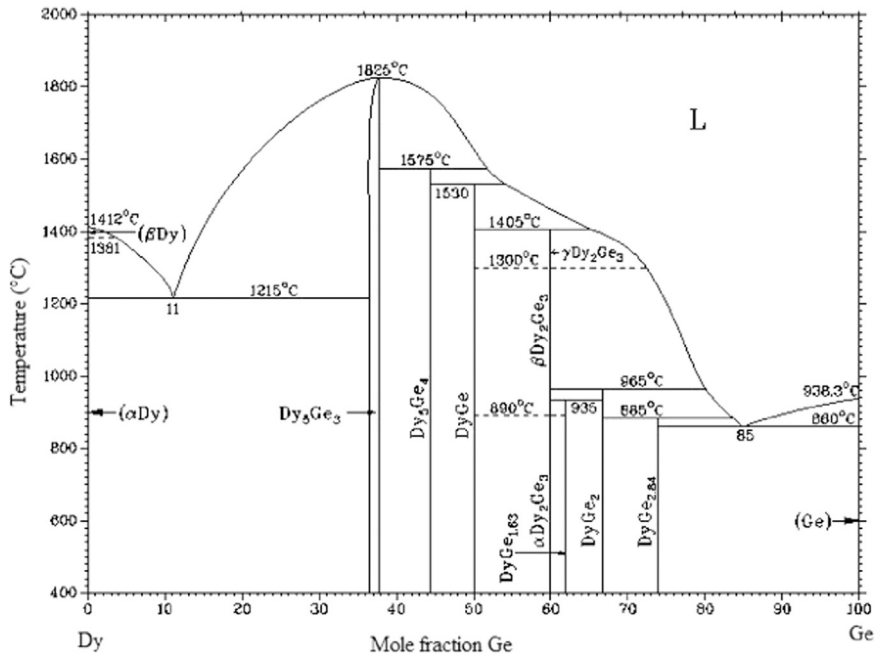

Fig. 2. Dysprosium-germanium phase diagram according to Eremenko et al. [16].

3. Procedure of thermodynamic modeling

In this section we will describe the approach of thermodynamic modeling of the studied systems. The optimization of the thermodynamic parameters for the Ge-Lu and Dy-Ge systems was performed using the PARROT module of the Thermo-Calc software developed by Jansson [27] and Sundman et al. [28], following a step-by-step optimization strategy.

As the first step we started with the liquid phase, using the zero order interaction parameters. Subsequently, intermetallic compounds were introduced into the modeling one by one. For these systems we first introduced the thermodynamic parameters of the congruent compounds optimized based of the data of the formation enthalpies given by Mishel and Klepa [18,24]. During the course of the optimization, the first and second order interaction parameters of the liquid phase were introduced to describe the shape of liquidus in the phase diagrams.

For Ge-Lu system, the liquid was treated as a substitutional solution using Redlich-Kister polynomial [9] in the first optimization, and then Kaptay model [10–12] was used in the second optimization in order to avoid the occurrence of an unwanted inverted miscibility gap in the solution phases at high temperatures during the calculation of the phase diagram.

For Dy-Ge system, the optimization was carried out in two steps. In the first treatment, Dy_5Ge_3 was assumed to be a stoichiometric compound. In the second, it was treated using the two-sublattice model [29,30]. The parameters obtained from the first treatment were used as starting values for the second.

For both systems the calculated invariant temperatures were adjusted in order to minimize the square sum of the errors between the experimental data and the computed values until most of the selected experimental information was reproduced within the expected uncertainty limits. The weights were adjusted systematically to improve the overall consistency of the data. The finally obtained thermodynamic parameters are listed in Tables 3–5.

4. Thermodynamic models

4.1. Pure elements

The Gibbs energy function for the pure element i(i = Lu, Dy, and Ge) in its stable state of the phase ϕ(ϕ = HCP_A3, BCC_A2, and DIAMOND_A4) is described by an equation of the following form:

$$G^{\Phi} - H^{\Phi}_{SER} = a + bT + cT\ln T + dT^2 + eT^3 + fT^{-1} + gT^7 + hT^{-9} \quad (1)$$

where $H_i^{SER}(298.15K)$ is the molar enthalpy of the element i at 298.15 K and 1 bar in its standard element reference (SER) state HCP_A3 for Lutetium and dysprosium and DIAMOND_A4 for Germanium, and T is the absolute temperature; a, b, c, …, h are the coefficients. The Gibbs energy of the element i, in its SER state, is denoted by *GHSERi* as follows:

$$GHSER_{Ge} = {}^0G_{Ge}^{Diamond,A4} = G_{Ge}^{Diamo,A4} - H_{Ge}^{Diamond,A4}(298.15K) \quad (2)$$

Table 1
Crystallographic data of the phases in Ge-Lu binary system.

Phase	at% Lu	Crystal system	Structure type	Space group	Pearson symbol	Reference
(Ge)	100	Cubic	C	$Fm\bar{3}m$	cF8	[13]
$Ge_{1.8}Lu$	35.7	Orthorhombic	Si_2Zr	*Cmcm*	oS12	[13,20]
αGe_3Lu_2	40	Hexagonal	AlB_2	*P6/mmm*	hP3	[37,38]
βGe_3Lu_2	40	……..	…	……	……	[37,38]
Ge_5Lu_4	44.4	……..	…	……	……	[37,38]
$Ge_{10}Lu_{11}$	52.3	Tetragonal	$Ge_{10}Ho_{11}$	I4/mmm	tI84	[13]
Ge_4Lu_5	55.5	Orthorhombic	Ge_4Sm_5	*Pnma*	oP36	[13]
Ge_3Lu_5	62.5	Hexagonal	Mn_5Si_3	$P6_3$/mcm	hP16	[39]
(Lu)	100	Hexagonal	Mg	*P63/mmc*	hP2	[13]

Table 2
Crystallographic data of the phases in Dy-Ge binary system.

Phase	at% Ge	Crystal system	Structure type	Space group	Pearson symbol	Reference
(αDy)	0	Hexagonal	Mg	*P63/mmc*	hP2	[40]
(βDy)	0	Hexagonal	W	$Im\bar{3}m$	cI2	[40]
Dy_5Ge_3	37.5	Hexagonal	Mn_5Si_3	$P6_3/mcm$	hp16	[40]
Dy_5Ge_4	44.4	Orthorhombic	Ge_4Sm_5	*Pnma*	oP36	[41]
DyGe	50	Orthorhombic	BCr	*Cmcm*	oS8	[41]
αDy_2Ge_3	60	Hexagonal	AlB_2	*P6/mmm*	hP3	[43]
βDy_2Ge_3	60	……	……	……	……	[43]
γDy_2Ge_3	60	……	……	……	……	[43]
$DyGe_{1.63}$	61.9	Orthorhombic	Ge_5Y_3	Fdd2	oF64	[41]
$DyGe_2$	66.7	Orthorhombic	Ge_2Tb	Cmmm	tI12	[41,42]
$DyGe_{2.84}$	73.9	Orthorhombic	$DyGe_3$	*Cmcm*	oS8	[41]
(Ge)	100	Cubic	C	$Fd\bar{3}m$	cF8	[40]

Table 3
Optimized thermodynamic parameters of the Ge-Lu system using Redlich–Kister equation [9].

Phases	Thermodynamic model	Parameters
Liquid	$(Ge,Lu)_1$	${}^0L^{Liq}_{Ge,Lu} = -319389.19 + 34.3270475*T$ ${}^1L^{Liq}_{Ge,Lu} = -8247.50556 + 2.10996002*T$ ${}^2L^{Liq}_{Ge,Lu} = 182281.853-65.6384780*T$
HCP_A3	$(Ge,Lu)_1(Va)^*_{.5}$	No excess term optimized
Diamond_A4	(Ge)	No excess term optimized
$Ge_{1.8}Lu$	$Ge_{.642}$:$Lu_{.358}$	$G^{Ge1.8Lu}_{Ge,Lu} - 0.642*^{298.15}H^{DIAMOND.A4}_{Ge} - 0.358*^{298.15}H^{HCP.A3}_{Lu}=$ $0.642*GHSERGe + 0.358*GHSERLu - 79{,}372.84 + 5.5101*T$
αGe_3Lu_2	$Ge_{.600}$:$Lu_{.400}$	$G^{\alpha Ge3Lu2}_{Ge,Lu} - 0.600*^{298.15}H^{DIAMOND.A4}_{Ge} - 0.400*^{298.15}H^{HCP.A3}_{Lu}=$ $0.600*GHSERGe + 0.400*GHSERLu - 81{,}269.45 + 1.9807*T$
βGe_3Lu_2	$Ge_{.600}$:$Lu_{.400}$	$G^{\beta Ge3Lu2}_{Ge,Lu} - 0.600*^{298.15}H^{DIAMOND.A4}_{Ge} - 0.400*^{298.15}H^{HCP.A3}_{Lu}=$ $0.600*GHSERGe + 0.400*GHSERLu - 81{,}266.43 + 1.9784*T$
Ge_5Lu_4	$Ge_{.556}$:$Lu_{.444}$	$G^{Ge5Lu4}_{Ge,Lu} - 0.556*^{298.15}H^{DIAMOND.A4}_{Ge} - 0.444*^{298.15}H^{HCP.A3}_{Lu}=$ $0.556*GHSERGe + 0.444*GHSERLu - 83{,}581.34 + 1.5371*T$
$Ge_{10}Lu_{11}$	$Ge_{.476}$:$Lu_{.524}$	$G^{Ge10Lu11}_{Ge,Lu} - 0.476*^{298.15}H^{DIAMOND.A4}_{Ge} - 0.524*^{298.15}H^{HCP.A3}_{Lu}=$ $0.476*GHSEGer + 0.524*GHSERLu - 87{,}780.92 + 0.7287*T$
Ge_4Lu_5	$Ge_{.444}$:$Lu_{.556}$	$G^{Ge4Lu5}_{Ge,Lu} - 0.444*^{298.15}H^{DIAMOND.A4}_{Ge} - 0.556*^{298.15}H^{HCP.A3}_{Lu}=$ $0.444*GHSERGe + 0.556*GHSERLu - 73{,}945.92 - 6.1449*T$
Ge_3Lu_5	$Ge_{.375}$:$Lu_{.625}$	$G^{Ge3Lu5}_{Ge,Lu} - 0.375*^{298.15}H^{DIAMOND.A4}_{Ge} - 0.625*^{298.15}H^{HCP.A3}_{Lu}=$ $0.375*GHSERGe + 0625*GHSERLu\text{-}\ 92{,}644.30 + 4.0872*T$

*(V_a) for Vacancy.

Table 4
Optimized thermodynamic parameters of the Ge-Lu system using Kaptay equation [10–12].

Phases	Thermodynamic model	Parameters
Liquid	$(Lu,Ge)_1$	${}^0L^{Liq}_{Ge,Lu} = -288557.01\exp(-5.6875E-05*T)$ ${}^1L^{Liq}_{Ge,Lu} = 1819.5035\exp(-2.4126E-04*T)$ ${}^2L^{Liq}_{Ge,Lu} = 90746.7571\exp(-1.2240E-04*T)$
HCP_A3	$(Ge,Lu)_1(Va)^*_{.5}$	No excess term optimized
Diamond_A4	(Ge)	No excess term optimized
$Ge_{1.8}Lu$	$Ge_{.642}$:$Lu_{.358}$	$G^{Ge1.8Lu}_{Ge,Lu} - 0.642*^{298.15}H^{DIAMOND.A4}_{Ge} - 0.358*^{298.15}H^{HCP.A3}_{Lu}=$ $0.642*GHSERGe + 0.358*GHSERLu - 77{,}119.0593 + 4.0568*T$
αGe_3Lu_2	$Ge_{.600}$:$Lu_{.400}$	$G^{\alpha Ge3Lu2}_{Ge,Lu} - 0.600*^{298.15}H^{DIAMOND.A4}_{Ge} - 0.400*^{298.15}H^{HCP.A3}_{Lu}=$ $0.600*GHSERGe + 0.400*GHSERLu - 81{,}159.2034 + 1.9801*T$
βGe_3Lu_2	$Ge_{.600}$:$Lu_{.400}$	$G^{\beta Ge3Lu2}_{Ge,Lu} - 0.600*^{298.15}H^{DIAMOND.A4}_{Ge} - 0.400*^{298.15}H^{HCP.A3}_{Lu}=$ $0.600*GHSERGe + 0.400*GHSERLu - 81{,}156.5143 + 1.9780*T$
Ge_5Lu_4	$Ge_{.556}$:$Lu_{.444}$	$G^{Ge5Lu4}_{Ge,Lu} - 0.556*^{298.15}H^{DIAMOND.A4}_{Ge} - 0.444*^{298.15}H^{HCP.A3}_{Lu}=$ $0.556*GHSERGe + 0.444*GHSERLu - 84{,}976.8792 + 2.00*T$
$Ge_{10}Lu_{11}$	$Ge_{.476}$:$Lu_{.524}$	$G^{Ge10Lu11}_{Ge,Lu} - 0.476*^{298.15}H^{DIAMOND.A4}_{Ge} - 0.524*^{298.15}H^{HCP.A3}_{Lu}=$ $0.476*GHSEGer + 0.524*GHSERLu - 89{,}767.9972 + 0.67871*T$
Ge_4Lu_5	$Ge_{.444}$:$Lu_{.556}$	$G^{Ge4Lu5}_{Ge,Lu} - 0.444*^{298.15}H^{DIAMOND.A4}_{Ge} - 0.556*^{298.15}H^{HCP.A3}_{Lu}=$ $0.444*GHSERGe + 0.556*GHSERLu - 72{,}026.5912–8.2441*T$
Ge_3Lu_5	$Ge_{.375}$:$Lu_{.625}$	$G^{Ge3Lu5}_{Ge,Lu} - 0.375*^{298.15}H^{DIAMOND.A4}_{Ge} - 0.625*^{298.15}H^{HCP.A3}_{Lu}=$ $0.375*GHSERGe + 0625*GHSERLu - 90{,}558.7991 + 2.0188*T$

* (V_a) for Vacancy.

Table 5
Comparison between the experimental invariant reactions [13] and the calculated ones (using Kaptay and Redlich-Kister equations) in the Ge-Lu system.

Reactions	Reaction Type	Literature [13]		This work Redlich-Kister[9]		This work Kaptay [10–12]	
		T/°C	x liq/at%Lu	T/°C	x liq/at%Lu	T/°C	x liq/at%Lu
L↔(Ge)	Melting	938.3	0.00	938.3	0.00	938.3	0.00
$L \leftrightarrow (Ge) + Ge_{1.8}Lu$	Eutectic	860	12	863	17.4	856	15.2
$\beta Ge_3Lu_2 \leftrightarrow L + Ge_{1.8}Lu$	Peritectic	1065	27.2	1060	26	1062	24.1
$Ge_{10}Lu_{11} \leftrightarrow L + \beta Ge_3Lu_2$	Peritectic	1330	37.7	1337	38	1332	35.6
$\beta Ge_3Lu_2 \leftrightarrow \alpha Ge_3Lu_2$	Polymorphic	1030	40	1031	40	1034	36.12
$Ge_5Lu_4 \leftrightarrow Ge_{10}Lu_{11} + \beta Ge_3Lu_2$	Peritectoid	1310	44.2	1310	44.2	1310	44.2
$Ge_4Lu_5 \leftrightarrow L + Ge_{10}Lu_{11}$	Peritectic	1720	50.2	1711	47.2	1719	47.09
$Ge_3Lu_5 \leftrightarrow L + Ge_4Lu_5$	Peritectic	1940	54.6	1939	54.3	1938	54.73
$Ge_4Lu_5 \leftrightarrow Ge_{10}Lu_{11} + Ge_3Lu_5$	Peritectoid	1660	55.5	1664	55.5	1652	55.5
$Ge_3Lu_5 \leftrightarrow L$	Congruent	2040	62.5	2044	62.5	2044	62.5
$L \leftrightarrow Ge_3Lu_5 + Lu$	Eutectic	1400	89	1398	86.5	1399	86.6
$L \leftrightarrow (Lu)$	Melting	1663	100	1663	100	1663	100

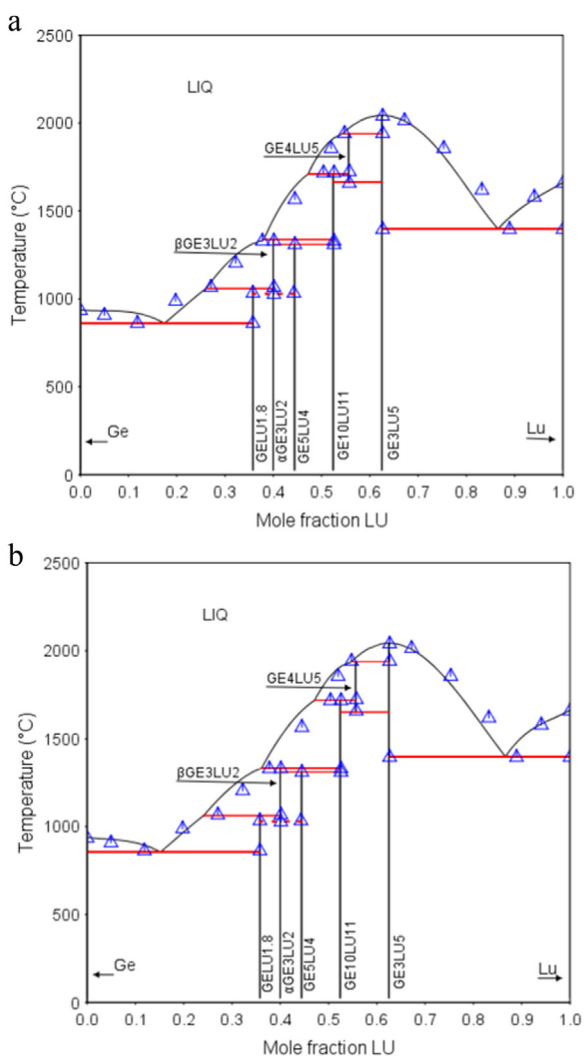

Fig. 3. a. Calculated Germanium-Lutetium phase diagram using the thermodynamic parameters optimized with the Redlich-Kister model [9] compared with the experimental data from the literature [13]. **b.** Calculated Germanium-Lutetium phase diagram using the thermodynamic parameters optimized with the Kaptay model [8–10] compared with the experimental data from the literature [13].

$$GHSER_{Lu} = {}^0G_{Lu}^{hcp.A3} = G_{Lu}^{hcp.A3} - H_{Lu}^{hcp.A3}(298.15K) \quad (3)$$

$$GHSER_{Dy} = {}^0G_{Dy}^{hcp.A3} = G_{Dy}^{hcp.A3} - H_{Dy}^{hcp.A3}(298.15K) \quad (4)$$

In the present modeling, the Gibbs energies functions are taken from the SGTE (Scientific Group Thermo data Europe) database compiled by Dinsdale [31].

4.2. Solution phases

The molar Gibbs energy of the solution phase ϕ including (liquid and HCP_A3 (αDy)) can be expressed as the sum of the reference part, the ideal part and the excess part:

$$G_m^{\phi} = {}^{ref}G_{\phi} + {}^{id}G_{\phi} + {}^{ex}G_{\phi} \quad (5)$$

$$G_m^{\phi} = x_{REE}\,{}^0G_{REE}^{\phi} + x_{Ge}\,{}^0G_{Ge}^{\phi} + RT(x_{REE}\ln x_{REE} + x_{Ge}\ln x_{Ge}) + {}^{ex}G_m^{\phi} \quad (6)$$

where R is the ideal gas constant, T is the temperature in Kelvin, x_{REE} and x_{Ge} are the mole fraction of the elements REE = (Lu or Dy) and Ge, respectively.

The excess Gibbs energy, ${}^{ex}G_m^{\phi}$ is described as follows:

$${}^{ex}G_m^{\phi} = x_{REE}x_{Ge}\sum_{i=0}^{n} {}^iL_{REE,Ge}^{\phi}(x_{REE} - x_{Ge})^i \quad (7)$$

The *ith* interaction parameters between components REE = (Lu or Dy) and Ge, ${}^iL_{REE,Ge}^{\phi}$ were temperature dependent are taken as:

a) Linear function of temperature for both binary systems Ge-Lu and Dy-Ge, with Redlich-Kister equation [9]

$${}^iL_{REE,Ge}^{\phi} = a_i + b_iT \quad (i = 0, 1, 2, \ldots\ldots) \quad (8)$$

with the excess entropy $s = -b_i$.

b) Exponential function of temperature only for Ge-Lu binary system. In order to avoid the occurrence of unwanted inverted miscibility gap at high temperature, Kaptay [10–12] suggested a new function to describe the excess Gibbs energy of the solution and liquid phases by assuming that the excess Gibbs energy should tend towards 0 at very high temperatures. This function is expressed as follow

$${}^iL_{Ge,Lu}^{liquid} = h_0^i\exp(-\frac{T}{\tau_0^i}) \quad (i = 0, 1, 2, \ldots\ldots) \quad (9)$$

with h_0^i the enthalpy part (J/mol) of ${}^iL_{Ge,Lu}^{liquid}$ at T = 0 K; the $\tau_0^i = \frac{h_0^i}{s_0^i}$ parameter must be always positive. By identifying h_0^i with a_i and τ_0^i with a_i/b_i, Eq. (9) becomes Eq. (10) with the excess entropy s = - b_i.

$${}^iL_{Ge,Lu}^{liquid} = a_i\exp((-\frac{b_i}{a_i})*T) \quad (10)$$

where the constants a_i and b_i are to be optimized in the present work.

Table 6
Comparison between the enthalpies and entropies of formation of the intermetallic compounds in the Ge-Lu system derived from the present optimization and taken from literature.

Intermetallic compounds	Enthalpies of formation J mol^{-1} at^{-1}	Entropies of formation J K^{-1} mol^{-1} at^{-1}	Technique Used	References
$Ge_{1.8}Lu$	− 79,372.8409	− 5.5101	Optimization[R-K]	Current work
	− 77,119.0593	− 4.0568	Optimization[K]	Current work
	− 87,800.03		Prediction	[33]
αGe_3Lu_2	− 81,269.4553	− 1.9807	Optimization[R-K]	Current work
	− 81,159.2034	− 1.9801	Optimization[K]	Current work
	− 88,766.84		Prediction	[33]
βGe_3Lu_2	− 81,266.4324	− 1.9784	Optimization[R-K]	Current work
	− 81,156.5143	− 1.9780	Optimization[K]	Current work
	− 84,766.84		Prediction	[33]
Ge_5Lu_4	− 83,581.3422	− 1.5371	Optimization[R-K]	Current work
	− 84,976.8792	− 2.00	Optimization[K]	Current work
	− 88,187.81		Prediction	[33]
$Ge_{10}Lu_{11}$	− 87,780.9250	− 0.7287	Optimization[R-K]	Current work
	− 89,767.9972	− 0.6787	Optimization[K]	Current work
	− 87,340.34		Prediction	[33]
Ge_4Lu_5	− 73,945.9234	6.1494	Optimization[R-K]	Current work
	− 72,026.5912	8.2441	Optimization[K]	Current work
	− 84,733.5		Prediction	[33]
Ge_3Lu_5	− 92,644.3046	− 4.0872	Optimization[R-K]	Current work
	− 90,558.7991	− 2.0188	Optimization[K]	Current work
	− 93,100 ± 2200		Calorimetry	[18]
	− 73,800 ± 2200		Calorimetry	[17]
	− 75,465.06		Prediction	[33]

[R–K] Redlich–Kister equation [9].
[K] Kaptay equation [10–12].

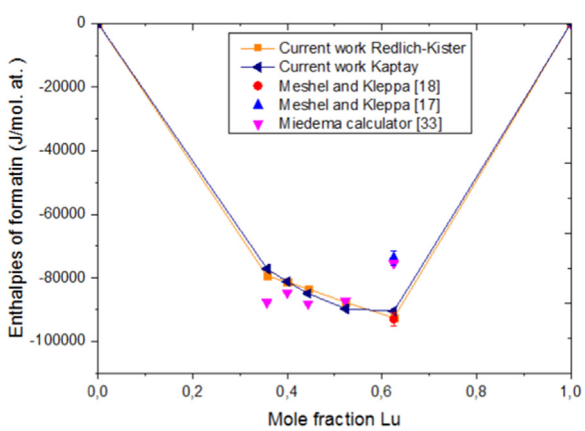

Fig. 4. Comparison between the calculated enthalpies of the intermetallic compounds using Redlich-Kister equation [9] (Orange line) and Kaptay equation [10–12] (blue line), with those measured by calorimetric method (blue triangle) [17] and (red circle) [18], and the predicted ones using Miedema model (pink triangle) [33] in Ge-Lu binary system. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.).

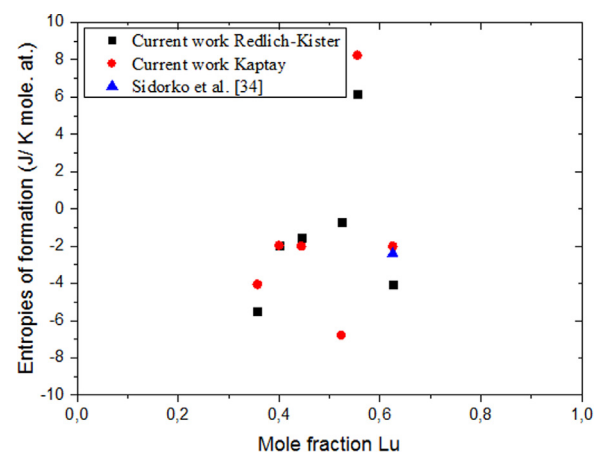

Fig. 5. Calculated entropies of formation of the intermetallic compounds with the Redlich-Kister () and Kaptay () models compared with the literature data () [34]. The reference states of pure elements Lutetium and Germanium are hcp_A3 and diamond_A4, respectively.

4.3. Stoichiometric compounds

All the intermetallic compounds in the (Dy or Lu)-Ge binary systems are treated as stoichiometric phases and described as A_pB_q except Dy_5Ge_3 phase in Dy-Ge phase diagram.

The Gibbs energy of each phase is denoted as ${}^0G_{REE_pGe_q}$ and described by the following formula

$$ {}^0G_{REE_pGe_q} = \frac{p}{p+q}{}^0G_{REE} + \frac{q}{p+q}{}^0G_{Ge} + a + bT \tag{11} $$

where ${}^0G_{REE}$ and ${}^0G_{Ge}$ are the Gibbs energy of the pure elements REE (Lutetium or Dysprosium) and Germanium, respectively; a and b are parameters to be determined for each of the compounds in the present work. The coefficients a can be estimated according to the corresponding experimental enthalpies of formation, then evaluated by taking into account both phase equilibria and thermodynamic data.

4.4. Intermediate phase with homogeneity range

The Dy_5Ge_3 phase is a non-stoichiometric compound which has a homogeneity range on the Dy-rich side of the strict stoichiometry Dy_5Ge_3. This phase was treated as the formula $(Dy)_{.625}$:$(Dy,Ge)_{.375}$ by a two-sublattices model [29,30]. The Gibbs energy function per mole of the formula unit is given by the following expression:

$$ \begin{aligned} {}^0G_m^{Dy5Ge3} &= G_m^{Dy5Ge3} - H^{SER}(298.15K) \\ &= y_{Ge}\,{}^0G_{Dy:Ge}^{Dy5Ge3} + y_{Dy}\,{}^0G_{Dy:dy}^{Dy5Ge3} + RT(y_{Dy}\ln y_{Dy} + y_{Ge}\ln y_{Ge}) \\ &\quad + y_{Dy}y_{Ge}\textstyle\sum_i {}^{\upsilon}L_{Dy:Dy,Ge}(y_{Dy} - y_{Ge})^{\upsilon} \end{aligned} \tag{12} $$

where y_{Dy} and y_{Ge} are the site fraction of component Dy and Ge located on two sublattices, and the parameter ${}^0G_{i:j}$ represents the Gibbs energy of the compound when the two sublattices are occupied by element i or

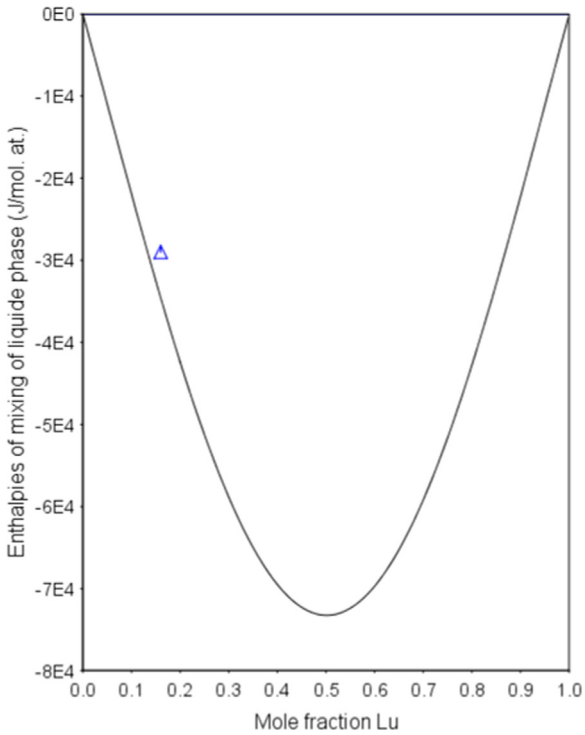

Fig. 6. Calculated integral enthalpies of formation of the liquid phase (continuous line) with the experimental data measured by Nikolaenko et al. [23] (open triangle) at T = 1275 K in Ge-Lu binary system.

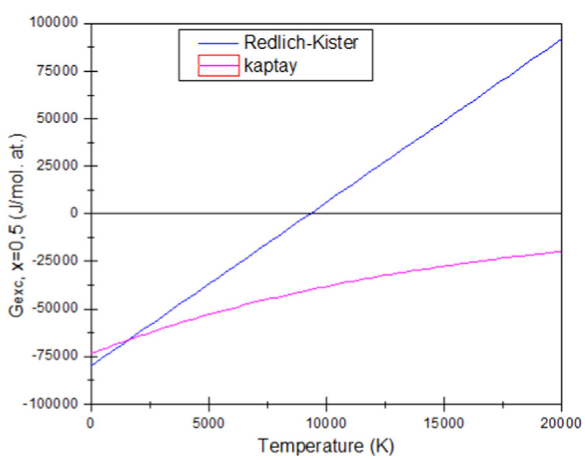

Fig. 7. Calculated excess Gibbs energies of the liquid phase in different temperature (0–20000 K) at X_{Ge} = 0.5 for Ge-Lu system using Redlich-Kister and Kaptay formalisms.

j (*i*, *j* = Dy, Ge). ${}^{\nu}L_{i:j}$ are the interaction parameters between Dy and Ge in the second or first sublattice, when the other sublattice is occupied by element *i* or *j*.

5. Results and discussion

5.1. The Ge-Lu binary system

Fig. 3**a** and **b** show the assessed phase diagrams obtained from the current thermodynamic modeling using Redlich-Kister (Fig. 3**a**) and Kaptay (Fig. 3**b**) formalisms along with the available experimental data of [13] superimposed.

As can be seen, the obtained phase diagram using Redlich-Kister formalism Fig. 3**a** agree well with the experimental data [13]. However,

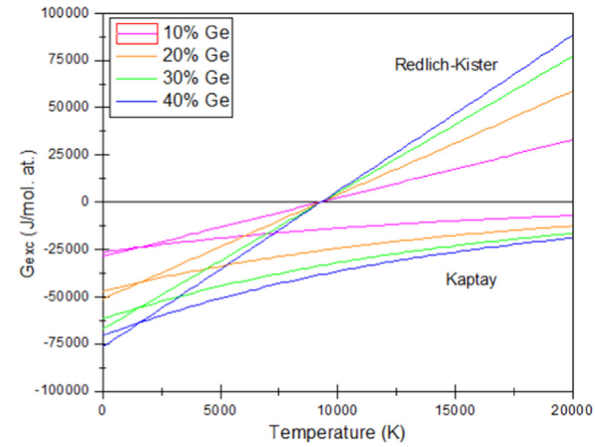

Fig. 8. Calculated excess Gibbs energies of the liquid phase in different temperature (0–20,000 K) and concentrations (X_{Ge} = 0.1, 0.2, 0.3, and 0.4) for Ge-Lu system using Redlich-Kister and Kaptay formalisms.

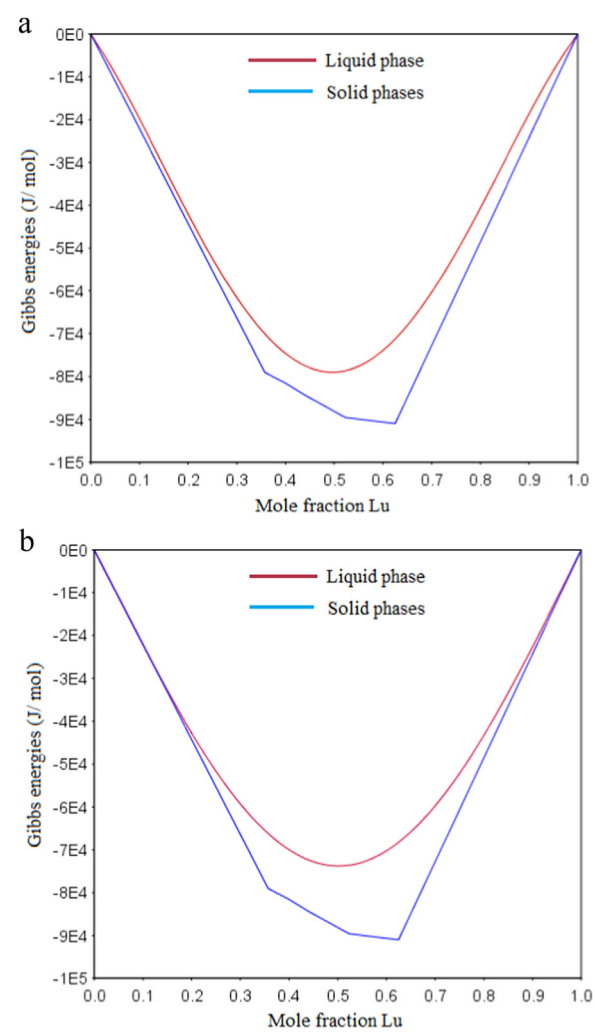

Fig. 9. b. Gibbs energies of the liquid phase and solid phases in the Ge-Lu system at T = 298.15 K using Redlich-Kister formalism. **b.** Gibbs energies of the liquid phase and solid phases in the Ge-Lu system at T = 298.15 K using Kaptay formalism.

exceptions are noted in the composition of the two calculated eutectic reactions $L \leftrightarrow Ge + Ge_{1.8}Lu$(17.4 at% Lu) and $L \leftrightarrow Ge_3Lu_5 + Lu$(86.5 at %Lu) show a small disagreement with experimental results (12 at% Lu)

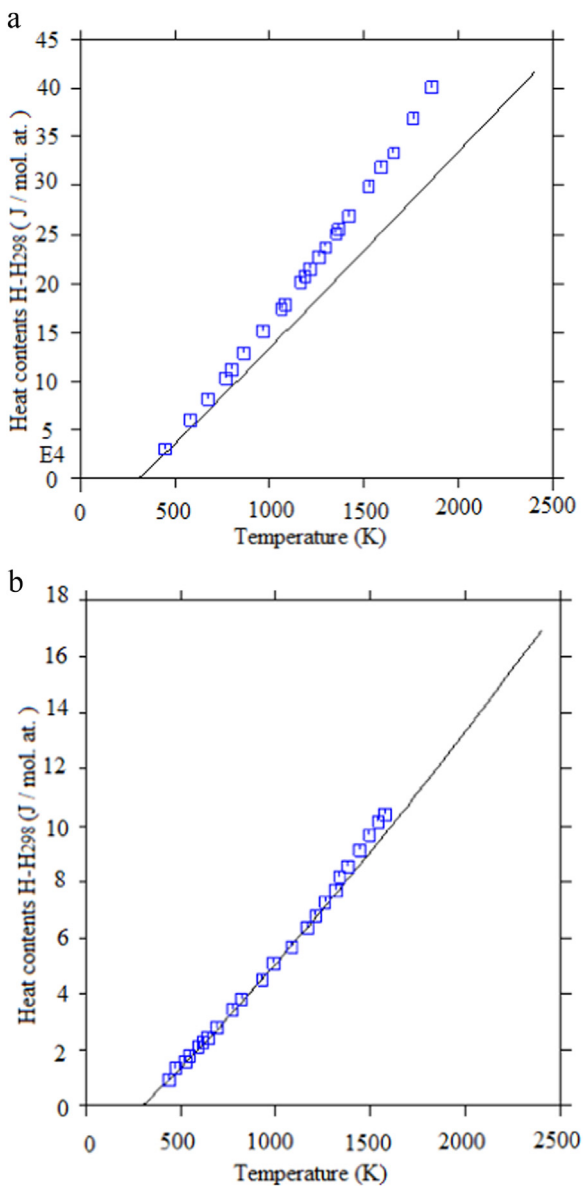

Fig. 10. a. Calculated heat contents for Ge_3Lu_5 along with the experimental values [19]. **b.** Calculated heat contents for Ge_3Lu_2 along with the experimental values [20].

and (89 at% Lu), respectively. We tried to minimize this difference and to obtain a better description of the liquid in the Lu and Ge rich parts by using the exponential temperature dependence given by Kaptay formalism. The achieved results are given in Fig. 3**b**. As can be seen the liquidus in the regions rich in Ge and Lu, and two eutectics points for the invariant reactions $L \leftrightarrow Ge + Ge_{1.8}Lu$ and $L \leftrightarrow Ge_3Lu_5 + Lu$ can be well reproduced. Small deviation can be detected on the liquidus linked tree peritectic reactions which were calculated in 47, 35.6, and 24.1 at % Lu, instead of 50.2, 37.7, and 27.2 at% Lu given by [13]. Which is caused by the symmetry of the liquidus and different weights given to the experimental data during the optimization. But our result makes the liquidus around the most stable compound more symmetric. Most of the calculated invariant reactions are within the uncertainty range (± 5 °C) compared with the measured data [13]. Only the temperatures of the invariant reaction $Ge_4Lu_5 \leftrightarrow Ge_{10}Lu_{11} + Ge_3Lu_5$ deviates about 8 °C in the case of Kaptay formalism from the corresponding measured data [13]. Such deviations are acceptable if the experimental errors are taken into consideration. The optimized model parameters for the Ge-Lu system are listed in Tables 3 and 4.

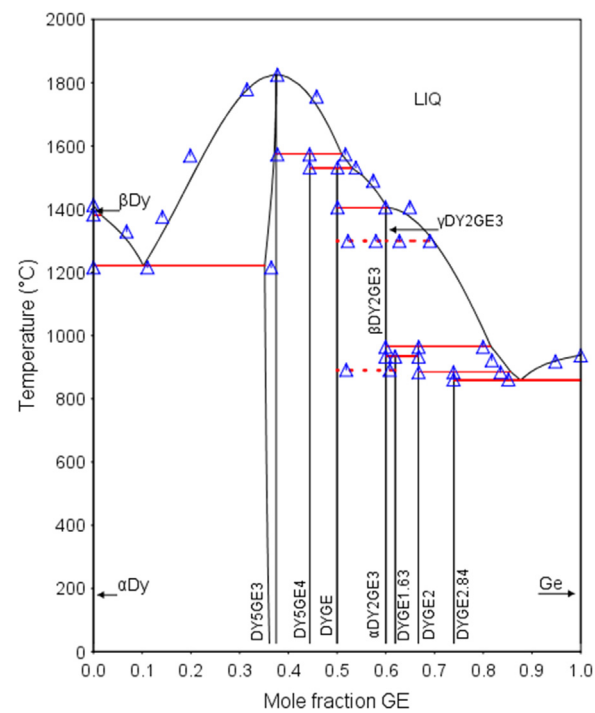

Fig. 11. Calculated Dysprosium-Germanium phase diagram using the thermodynamic parameters optimized with the Redlich-Kister model [9] compared with the experimental data from the literature [16].

Table 5 displays the calculated invariant temperatures along with the experimental ones.

The calculated phase boundaries are in good agreement for all the experimental data and for the two formalisms used in this work. Especially, no liquid miscibility gap was observed within the defined temperature range of the SGTE database (298.15–6000 K) [32] in the present calculation for both descriptions of the liquid phase. The formation enthalpies for different intermetallic phases at 25 °C were further calculated in comparison with the reported values by experimental measurements [17,18] as well as the predicted ones using Miedema model [33]. As shown in Table 6 and Fig. 4, our assessed results using two formalisms agree well with the experimental data given in ref [18] for Ge_3Lu_5 phase. The calculated entropies of formation of six intermetallic compounds are plotted in Fig. 5 based on the present assessed thermodynamic parameters. A good agreement is noted between our calculated value and the data from [34] for Ge_3Lu_5 compound.

Fig. 6 shows the mixing enthalpy of liquid phase in Ge-Lu system calculated in this work over the whole range of composition compared with the calorimetric result reported by Nikolaenko et al. [23] at 1275 K for a composition of 16%Lu. The agreement with experimental measurements is satisfactory.

Taking into account the equations given in Tables 3 and 4 for °L values:

$$Redlich - Kister: \Delta G^E_{x=0.5} = 0.25(-319389.19 + 34.32*T)$$

$$Kaptay: \Delta G^E_{x=0.5} = 0.25(-288557\exp(-5.68E-05*T))$$

The excess Gibbs energies of mixing of the liquid phase at $X_{Ge} = 0.5$ (Fig. 7) and $X_{Ge} = 0.1$, 0.2, 0.3, and 0.4 (Fig. 8) were plotted as function of temperature from 0 to 20,000 K for the two models (Redlich-Kister and Kaptay).

i. At low temperatures Redlich-Kister polynomial provides more negative excess Gibbs energy compared to the Kaptay formalism.
ii. At medium temperatures Kaptay formalism provides more negative excess Gibbs energy compared to the Redlich-Kister polynomial.

Table 7
Optimized thermodynamic parameters of the Dy-Ge system using Redlich–Kister equation [9].

Phases	Thermodynamic model	Parameters
Liquid	$(Dy,Ge)_1$	${}^0L^{Liq}_{Dy,Ge} = -268121.57-1.1442*T$
		${}^1L^{Liq}_{Dy,Ge} = 37646.42-26.70302*T$
		${}^2L^{Liq}_{Dy,Ge} = 54703.93 + 9.9947*T$
HCP_A3	$(Dy,Ge)_1(Va)_{.5}$*	${}^1L^{HCP.A3}_{Dy,Ge} = -69413.23 - 20.4258*T$
BCC_A2	$(Dy,Ge)_1(Va)_3$*	No excess term optimized
Diamond_A4	(Ge)	No excess term optimized
Dy_5Ge_3	$(Dy)_{.625}$:$(Dy,Ge)_{.375}$	$G^{Dy5Ge3}_{Dy,Ge} - 0.625*{}^{298.15}H^{HCP.A3}_{Dy} - 0.375*{}^{298.15}H^{DIAMOND.A4}_{Ge}=$
		$0.625*GHSERDy + 0.375*GHSERGe - 92{,}218.67 + 2.2256*T$
		$G^{Dy5Ge3}_{Dy,Dy} - H^{DIAMOND.A4}_{Ge} = 25000 + GHSERDy + 1000 + 8*T$
		$L^{Dy5Ge3}_{Dy:Dy,Ge} = -27000 - 1*T$
Dy_5Ge_4	$Dy_{.556}$:$Ge_{.444}$	$G^{Dy5Ge4}_{Dy,Ge} - 0.556*{}^{298.15}H^{HCP.A3}_{Dy} - 0.444*{}^{298.15}H^{DIAMOND.A4}_{Ge}=$
		$0.556*GHSERDy + 0.444*GHSERGe - 91{,}066.89 + 1.4098*T$
DyGe	$Dy_{.5}$:$Ge_{.5}$	$G^{DyGe}_{Dy,Ge} - 0.5*{}^{298.15}H^{HCP.A3}_{Dy} - 0.5*{}^{298.15}H^{DIAMOND.A4}_{Ge}=$
		$0.5*GHSERDy + 0.5*GHSERGe - 90{,}183.99 + 1.2664*T$
αDy_2Ge_3	$Dy_{.4}$:$Ge_{.6}$	$G^{\alpha Dy2Ge3}_{Dy,Ge} - 0.4*{}^{298.15}H^{HCP.A3}_{Dy} - 0.6*{}^{298.15}H^{DIAMOND.A4}_{Ge}=$
		$0.4*GHSERDy + 0.6*GHSERGe - 85{,}929.65 + 2.5446*T$
βDy_2Ge_3	$Dy_{.4}$:$Ge_{.6}$	$G^{\beta Dy2Ge3}_{Dy,Ge} - 0.4*{}^{298.15}H^{HCP.A3}_{Dy} - 0.6*{}^{298.15}H^{DIAMOND.A4}_{Ge}=$
		$0.4*GHSERDy + 0.6*GHSERGe - 85{,}880.07 + 2.5020*T$
γDy_2Ge_3	$Dy_{.4}$:$Ge_{.6}$	$G^{\gamma Dy2Ge3}_{Dy,Ge} - 0.4*{}^{298.15}H^{HCP.A3}_{Dy} - 0.6*{}^{298.15}H^{DIAMOND.A4}_{Ge}=$
		$0.4*GHSEDy + 0.6*GHSERGe - 85{,}876.66 + 2.4998*T$
$DyGe_{1.63}$	$Dy_{.380}$:$Ge_{.619}$	$G^{DyGe1.63}_{Dy,Ge} - 0.380*{}^{298.15}H^{HCP.A3}_{Dy} - 0.619*{}^{298.15}H^{DIAMOND.A4}_{Ge}=$
		$0.380*GHSERDy + 0.619*GHSERGe - 85{,}082.51 + 4.9393*T$
$DyGe_2$	$Dy_{.333}$:$Ge_{.667}$	$G^{DyGe2}_{Dy,Ge} - 0.333*{}^{298.15}H^{HCP.A3}_{Dy} - 0.667*{}^{298.15}H^{DIAMOND.A4}_{Ge}=$
		$0.333*GHSERDy + 0.667*GHSERGe - 81{,}531.45 + 9.2563*T$
$DyGe_{2.84}$	$Dy_{.260}$:$Ge_{.739}$	$G^{DyGe2.84}_{Dy,Ge} - 0.260*{}^{298.15}H^{HCP.A3}_{Dy} - 0.739*{}^{298.15}H^{DIAMOND.A4}_{Ge}=$
		$0.260*GHSERDy + 0.739*GHSERGe - 66{,}031.19 + 8.9522*T$

* (V_a) for Vacancy.

Table 8
Comparison between the experimental invariant reactions [16] and the calculated ones using Redlich-Kister equation in the Dy-Ge system.

Reactions	Reaction Type	Literature [16]		This work Redlich-Kister [9]	
		T/°C	xliq/ at%Ge	T/°C	xliq/at %Ge
L↔(Dy)	Melting	1412	0	1412	0.00
(αDy)↔ (βDy)	Polymorphic	1381	0	1381	0.00
$L \leftrightarrow (\alpha Dy) + Dy_5Ge_3$	Eutectic	1215	11	1214	10.4
$L \leftrightarrow Dy_5Ge_3$	Congruent	1825	37.7	1825	37.5
$Dy_5Ge_3 \leftrightarrow L + Dy_5Ge_4$	Peritectic	1575	51.6	1575	50.9
$Dy_5Ge_4 \leftrightarrow L + DyGe$	Peritectic	1530	54	1530	53
$DyGe \leftrightarrow L + \gamma Dy_2Ge_3$	Peritectic	1405	64.9	1405	60.2
$\alpha Dy_2Ge_3 \leftrightarrow \beta Dy_2Ge_3$	Polymorphic	890	60	890	60
$\beta Dy_2Ge_3 \leftrightarrow \gamma Dy_2Ge_3$	Polymorphic	1300	60	1300	60
$DyGe_{1.63} \leftrightarrow DyGe_2 + \beta Dy_2Ge_3$	Peritectoid	935	61.8	935	61.8
$\beta Dy_2Ge_3 \leftrightarrow L + DyGe_2$	Peritectic	965	79.9	966	81.5
$DyGe_2 \leftrightarrow L + Dy_1Ge_{2.84}$	Peritectic	885	83.5	885	85.6
$L \leftrightarrow DyGe_{2.84} + Ge$	Eutectic	860	85.1	859	87.6
$(Ge) \leftrightarrow L$	Melting	938.3	100	938.3	100

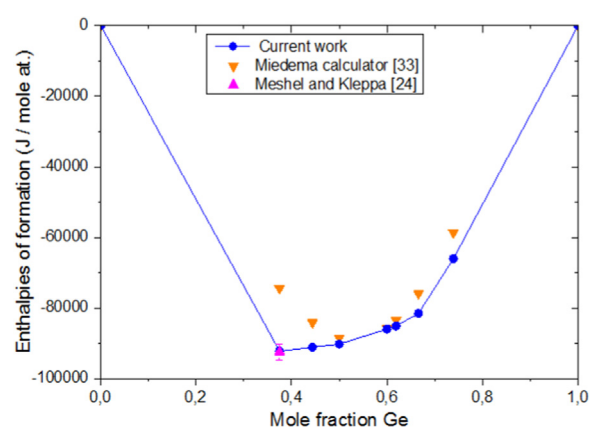

Fig. 12. Comparison between the calculated enthalpies of the intermetallic compounds using Redlich-Kister equation [9] (blue line) with those measured by calorimetric method (pink triangle) [24], and the predicted ones using Miedema model (orange triangle) [33] in Dy-Ge binary system. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.).

iii. At high temperatures the excess Gibbs energy behaves in a different manner. For Redlich-Kister model, this function changes its sign from negative to positive and goes to more and more positive values with higher temperatures, which is probably unphysical [44]. In contrary, for Kaptay model the excess Gibbs energy tends towards zero at high temperatures, in accordance with the general rule [44].

In Fig. 9**a** and **b** we checked the liquid re-stabilization at low temperatures for both formalisms (Redlich-Kister and Kaptay, respectively) by comparing the Gibbs energy of solid phases (broken lines) to the Gibbs energy of the liquid phase (continuous lines) at 298.15 K. As claimed by Schmid-Fetzer et al. [45], the low-T artefact has a high probability to appear at lower T-s, but the unary data are known only above T = 298.15 K [31]. Then the Gibbs energies of solid phases will be compared to the Gibbs energy of the liquid phase at this temperature. One can see that the Gibbs energy line for the liquid phase is situated in a more positive range compared to the Gibbs energy of stable solid phases in both cases. Even if in the case of Kaptay formalism

Table 9
Comparison between the enthalpies of formation of the intermetallic compounds in the Dy-Ge system derived from the present optimization and taken from literature.

Intermetallic compounds	Enthalpy of formation J mol^{-1} at^{-1}	Entropy of formation J mol^{-1} at^{-1} K^{-1}	Technique Used	References
Dy_5Ge_3	− 92,400 ± 2300	− 2.22	Calorimeter	[24]
	− 74,464.86		Prediction	[33]
	− 92,218.67		Optimization[R-K]	Current work
Dy_5Ge_4	− 84,111.58	− 1.40	Prediction	[33]
	− 91,066.89		Optimization[R-K]	Current work
DyGe	− 88,369.35	− 1.26	Prediction	[33]
	− 90,183.99		Optimization[R-K]	Current work
αDy_2Ge_3	− 85,718.09	− 2.54	Prediction	[33]
	− 85,929.65		Optimization[R-K]	Current work
βDy_2Ge_3	− 85,718.09	− 2.50	Prediction	[33]
	− 85,880.07		Optimization[R-K]	Current work
γDy_2Ge_3	− 85,718.09	− 2.49	Prediction	[33]
	− 85,876.66		Optimization[R-K]	Current work
$DyGe_{1.63}$	− 83,472.15	− 4.93	Prediction	[33]
	− 85,082.51		Optimization[R-K]	Current work
$DyGe_2$	− 75,917.95	− 9.25	Prediction	[33]
	− 81,531.45		Optimization[R-K]	Current work
$DyGe_{2.84}$	− 58,672.51	− 8.95	Prediction	[33]
	− 66,031.19		Optimization[R-K]	Current work

[R–K]Redlich–Kister equation [9].

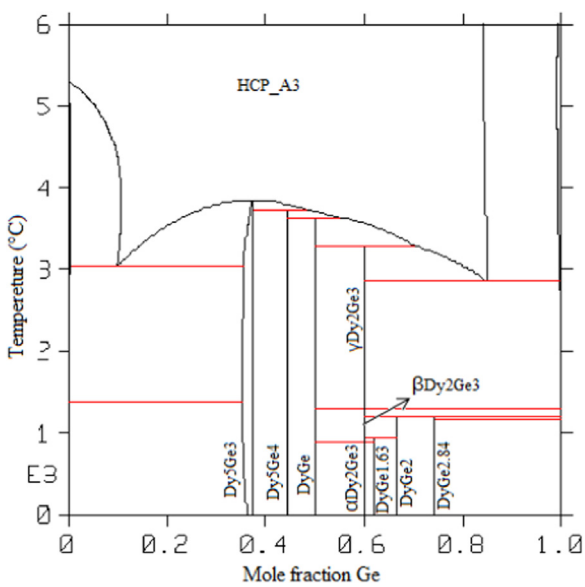

Fig. 13. Calculated Dysprosium–Germanium phase diagram with the Redlich–Kister equation [9] when the liquid phase is suspended.

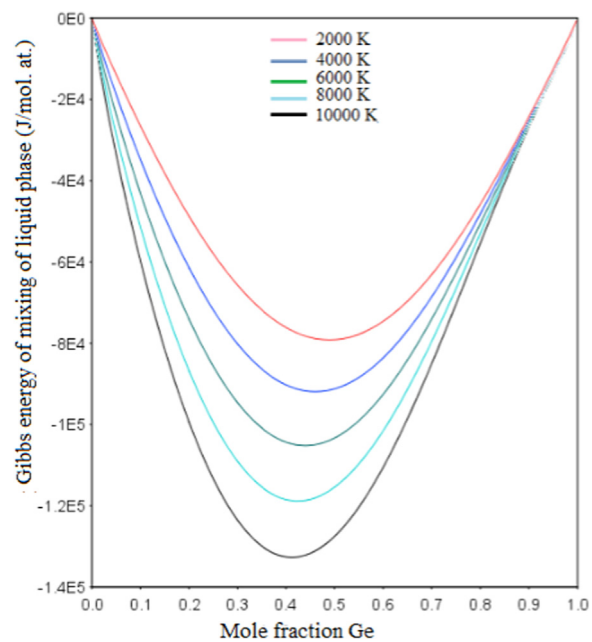

Fig. 14. Calculated Gibbs free energy of mixing of the liquid phase at different temperatures (2000–10,000 K) using the Redlich–Kister model [9].

the Gibbs energies of liquid and solid phases are too close at T = 298.15 K in the Lu and Ge rich parts, one can conclude that the low-T artefact does not take place in this system. Moreover, when we plot the phase diagrams starting from T = 298 K, we can't see any re-stabilization of the liquid phase below the solidus, this conclusion is in agreement with the results proven recently by Kaptay [46].

The heat content changes also called the enthalpy increment H-H298.15 are shown graphically as a function of temperature for Ge_3Lu_5 and Ge_3lu_2 in Fig. 10**a** and **b**, respectively. The calculated heat contents increase monotonically in the range 300–2355 K for Lu_5Ge_3, in accordance with the measured data in the literature [19], and in the range from 442 to 1576 K for Ge_3Lu_2 in accordance with the experimental data given in ref [20]. The comparison between the measured heat contents and the calculated values is fairly good, even though the measured enthalpies were not included during the optimization procedure. The small anomalies observed with the experimental data were may be caused by the reactions involved between these two compounds and the others in the phase equilibria.

5.2. The Dy-Ge binary system

The calculated Dy–Ge phase diagram compared with experimental data of Eremenko et al. [16] is given in Fig. 11. A good agreement is achieved for both invariant temperatures and compositions compared with the results of ref [16]. Table 7 presents the finally evaluated thermodynamic parameters of each phase in the Dy-Ge system. The comparison between measured and computed invariants equilibria are given in Table 8.

Fig. 12 and Table 9 present the calculated enthalpies of formation at 298.15 K with the standard element reference state compared with experimental data [24] and predicted values from the semiempirical model of Miedema [33]. The present calculation agrees well with the data of Meschel and Kleppa [24] for Dy_5Ge_3 however Miedema model gives less exothermic value for Dy_5Ge_3 phase.

In Fig. 13 we plot the phase diagram when the liquid phase is

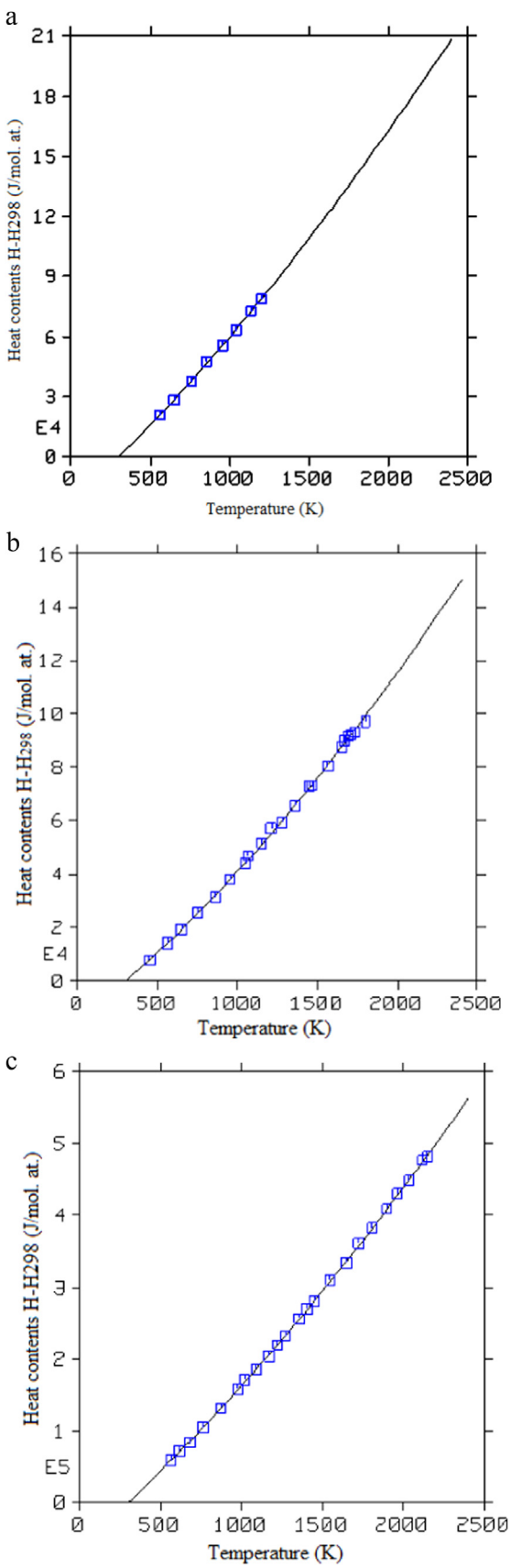

Fig. 15. a. Calculated heat contents for $DyGe_2$ along with the experimental values [25]. **b.** Calculated heat contents for DyGe along with the experimental values [25]. **c.** Calculated heat contents for Dy_5Ge_3 along with the experimental values [25].

suspended. We note that when the liquid phase is not considered during the calculation of the phase diagram at high temperature, the stoichiometric phases disappear and only the hcp_A3 solid solution is calculated on the whole germanium composition range. As indicated in Refs. [35,36] our optimized parameters are reliable.

In Fig. 14, the Gibbs energy of mixing of the liquid phase was calculated at different temperatures: 2000–10,000 K with Redlich–Kister model for Dy-Ge system. This figure shows that the Gibbs energies of mixing for the liquid phase become more negative when the temperature increases. The liquid is stable at very high temperatures, no spinodal points were detected until a temperature of 10,000 K using the Redlich–Kister model. We mention that no formation of an inverted miscibility gap was detected at very high temperatures, in the region of validity of parameters, during the optimization.

This conclusion is consistent with the condition given in ref [10]. Using Redlich-Kister formalism at high temperature $|A^0_{Dy,Ge}| << |B^0_{Dy,Ge}T|$, the inverted miscibility gap arise for n = 0 when $B^0_{Dy,Ge} > 2R$ $(S^0_{Dy,Ge} < -2R)$ [10], where n is the zero power, R is the ideal gas constant, and $B^0_{Dy,Ge} = -S^0_{Dy,Ge}$. In the case of Dy-Ge system, $B^0_{Dy,Ge}(=-1.14J/mol.\ K) < 2R(=16.62J/mol.\ K)$ so no miscibility gaps arises using Redlich Kister model for this system.

Fig. 15**a, b**, and **c** present, respectively, the comparison between the computed results and the experimental data of the enthalpies H-H298.15 of the compounds Dy_5Ge_3, DyGe, and $DyGe_2$ at different temperatures [25]. It can be seen that a good agreement between the calculated and the experimental data has been achieved.

6. Conclusion

The modeled Gibbs energy function of each solution phase and intermetallic compounds in the Ge-Lu and Dy-Ge were used to simultaneously construct the phase diagrams and validate the thermodynamic properties of these binary systems. For both systems the calculated phase diagrams and thermodynamic quantities have been reproduced reasonably and agree well with the available experimental data in the literature. Concerning Ge-Lu system the two models of the temperature dependence of the excess terms of the solution phases give nearly similar calculated results. To improve the present assessments of both systems, more experimental work on the phase diagram and thermodynamic properties such as the integral and partial enthalpies of mixing of the liquid phase as well as the enthalpies of formation of the compounds are needed.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at http://dx.doi.org/10.1016/j.calphad.2018.05.001.

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